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Enhancing Public Safety: Real-Time Violence Detection and Notification System

Rupsa Rupa Ojha, Harshita Chawdary, Surbhi Saraswat*[1]

UPES, Dehradun, 248001, India

Abstract

The increasing incidence of violent events in public and private spaces necessitates the development of advanced security measures. This work introduces a novel Real-Time Violence Detection and Notification System, leveraging Faster R-CNN and MobileNetV2 architectures to analyze surveillance footage effectively. It enhances public safety by providing real-time alerts to security personnel and authorities via a Telegram bot, ensuring swift and efficient intervention. The system utilizes a combination of machine learning algorithms and computer vision techniques to accurately identify aggressive behavior while distinguishing it from non-violent activities. Comprehensive evaluations highlight the system's high accuracy, and a comparative analysis demonstrates superior performance over existing models in detecting violent behavior.

Keywords: Alert generation, CNN, Faster RNN, MobilenetV2, Real-time violence detection.

1. Introduction

The increasing incidents of violent behavior in communities are viewed as a serious threat to public safety, impacting not just productivity but also property values and social services. Effective intervention relies on the real-time detection and identification of violent activities, which requires continuous analysis of video feeds from numerous surveillance cameras. Public video surveillance systems are widely used globally, providing critical information for security applications. However, the monitoring of footage manually is considered impractical and inefficient. Therefore, the automatic detection of violent scenes is deemed crucial to prevent crime and violence, enabling prompt responses to incidents by authorities.

[1] * Corresponding author. Tel.: +91 8127705936
E-mail address: Surbhi.saraswat@ddn.upes.ac.in

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Deep learning methods, particularly Convolutional Neural Networks (CNNs) [9], have shown promise in the extraction of spatiotemporal features from video footage. A Real-Time Violence Detection and Notification System utilizing the MobileNetV2 architecture is introduced in this work, which balances high accuracy with computational efficiency [11]. The proposed method involves the identification of human presence in video frames, the extraction of frames with predicted violent activities, and the discarding of irrelevant ones. The relevant frames are saved and enhanced. Alerts, including images, timestamps, and locations, are sent to authorities via a Telegram bot, ensuring immediate and informed responses. The contributions in this work include the integration of a real-time detection algorithm, the development of a comprehensive alert system, and comparative analysis with existing models and algorithms, demonstrating improved detection accuracy and reduced false positive predictions.

The major contributions of this work are as follows:

- We propose an end-to-end system that aims to detect violence from surveillance videos in real-time and generate alerts to notify the authorities to ensure timely intervention.
- We validate the system across multiple available datasets and analyze the effects of various model parameters to achieve optimal performance. The system is further evaluated on newly collected real-time data.
- We also compare the performance of the system with other existing systems and the results indicate that the proposed system outperforms other systems in terms of accuracy and notification delays.

The rest of the paper is organized as follows: In the next section, we present a review of relevant background information and existing approaches to this problem. Section 3 presents the proposed methodology and system design. We then present the results of our experiments and discuss their implications in Section 4. Finally, Section 5 summarizes our findings and concludes the paper.

2. Related Work

There has been an increasing interest in the recent past in the utilization of automated systems for violence detection to improve safety in various environments. This section presents some relevant work on violence detection and notification with a focus on their approaches, advantages, and limitations.

2.1. Violence detection

MobileNetV2 is a lightweight CNN architecture that has been applied in numerous real time image and video processing applications. Numerous studies have explored different methodologies and datasets. For instance, Sandler et al. (2018) [2] demonstrated the efficiency of MobileNetV2 in real-time applications, highlighting its performance on mobile and embedded platforms.

A comprehensive dataset called XD-Violence was introduced by Peng Wu et al. (2020) [4], consisting of 4,754 untrimmed videos with accompanying audio. A multimodal neural network was developed that incorporates three parallel branches to capture different relationships among video snippets. Visual and audio features are effectively combined by this model, enhancing the detection of violent actions. However, the requirement for extensive computation time by the model limits its application in real-time scenarios, presenting a challenge for practical deployment.

A deep learning model known as ViolenceNet was proposed by Fernando J. Rendón-Segador and his team (2021) [5], which integrates a modified DenseNet architecture with multi-head self-attention and bidirectional long short-term memory (LSTM) networks. High accuracy across various datasets was achieved by this combination, demonstrating the potential of attention mechanisms in improving performance. Despite its strengths, generalization across different datasets was struggled with by the model, indicating a need for further refinement in feature learning.

MobileNetV2 was applied by Phat Nguyen Huu et al. (2022) [7] to achieve a detection accuracy of 99.37% for both masked and non-masked faces. Retina Face with ResNet50 is used for initial detection by their system, which operates at a speed of 3 frames per second (fps), targeting environments like schools and offices. Future enhancements that were discussed include the improvement of performance on blurred faces and the optimization of the model for faster processing in embedded systems, which is considered crucial for real-time applications. A lightweight yet effective model for violence detection in surveillance footage was proposed by Vijeikis et al. (2022) [8]. A U-Net-like architecture with MobileNetV2 for spatial feature extraction is employed by this model, complemented by LSTM for temporal analysis. The model was tested on datasets such as RWF-2000, Movie Fights, and Hockey Fights, achieving an average accuracy of 82% and precision of 81%. These results indicate that its suitability for deployment on low-power edge devices is established, making it a viable option for real-time surveillance in various contexts.

2.2. Real-time notification

An effective violence detection system must be capable of immediately informing the authorities once violence is positively detected. Some earlier work in the field of violence detection has also examined several formats of notification, including the simplest forms of notification, SMS and email notification, as well as more complicated forms of notification through mobile applications. A smartphone application was developed by Kim, Seul-Kee et al. (2016) [1] that sent push notifications to consumers regarding occurrences that had been flagged. However, the main drawback of their system was that it relied on push notifications, which would be of little use if the smartphone could not be accessed quickly by the user.

The utilization of messaging tools like Telegram has not been discussed much; however, it is considered quite promising. Telegram bots, due to their user-friendly interface and high level of usage, are seen as a viable option for real-time notification. Thus, by connecting with Telegram, violence detection systems can guarantee that notifications will be sent to multiple users, including security specialists and police officers.

3. Proposed Methodology

This section outlines the methodology employed for real-time violence detection, followed by a discussion of the alert module developed for generating and sending real-time notifications. As illustrated in Fig. 1, the system architecture is comprised of four major stages: data collection and preprocessing, model design and training, violence detection module, and alert module.

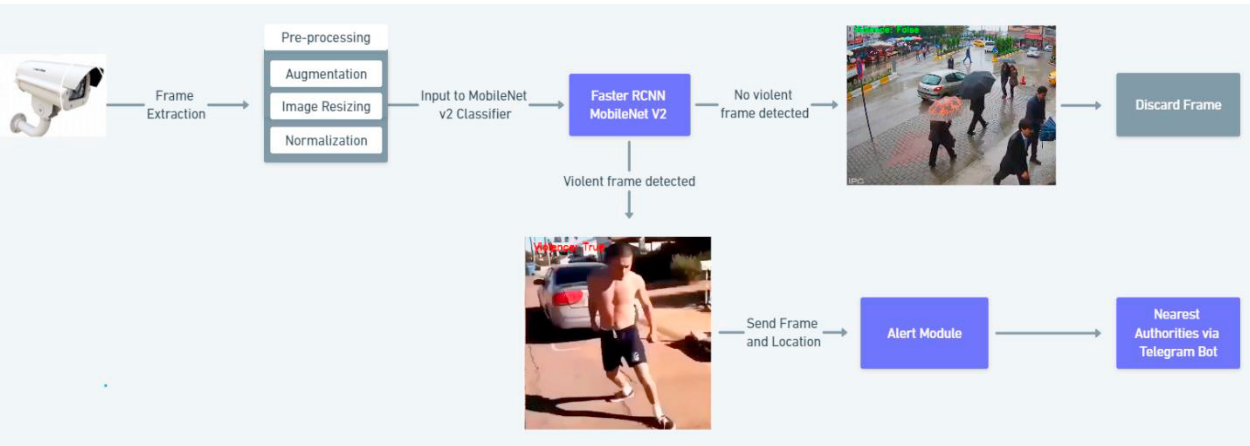

Fig 1. Architecture of Violence detection and notification system

The system was tested using video footage from public spaces. The experimental setup consisted of a high-performance processing unit for model training, two datasets consisting of both balanced and unbalanced data, and the TensorFlow framework as the primary framework. Telegram port configuration was used to ensure that real-time video streams were effectively managed. The evaluation metrics used included precision, recall, and F-1 score to assess model performance.

3.1. Data Collection and Preprocessing:

Surveillance video clips are taken as input for evaluation by the system. The first step in preparing the data involves the decomposition of the video into individual frames using the cv2.VideoCapture library. This process ensures that temporal information is retained while enabling detailed inspection of each frame. An optimized video frame rate is determined to ensure consistent sampling.

Each frame then undergoes several preprocessing steps to enhance its suitability for analysis:

- **Augmentation:** Frames are augmented using techniques such as flipping, zooming, random brightness adjustment, and rotation to increase dataset variability and mitigate duplication. Augmented frames are converted from BGR to RGB color space and resized to a fixed dimension of 128x128 pixels to ensure uniformity. The augmentation process employs the `imgaug` library, while `imageio` handles image input and output operations.
- **Enhancement:** Following augmentation, frames are enhanced using techniques such as noise reduction, contrast adjustment, and sharpening. These enhancements improve clarity and facilitate the accurate detection of violent activities by making key features more discernible.
- **Filtering:** To optimize computational resources, selective filtering is applied to focus on frames with a higher likelihood of depicting violent behavior. Non-violent frames are discarded, concentrating resources on the most relevant data.

3.2. Model design and training

After the frames are preprocessed, they are input to the MobilenetV2 model for training and validation. The model design is presented ahead discussing its different layers, activation function and loss function.

- **Depthwise Separable Convolution:** It is an efficient way to perform convolution operations in neural networks. Instead of using a single large filter, it breaks down the process into two smaller steps: Depthwise Convolution and pointwise convolution. In the first step, each input channel is processed independently using a single filter. This reduces the number of calculations. The outputs from the depthwise convolution are then combined using 1x1 filters. This allows the network to learn relationships between the channels. This enables Depthwise Separable Convolution to achieve high accuracy while using fewer computational resources.
- **Residual Connections:** Linear bottlenecks and inverted residuals are used to increase performance. These procedures help preserve information and reduce the memory usage of the network by storing minimal data. Linear bottlenecks decrease the dimensionality of feature maps, which helps reduce the number of parameters and computational load while maintaining essential information. Inverted residuals expand feature maps prior to depthwise separable convolutions, enabling the effective capture of complex features while preserving computational efficiency.
- **Activation function and loss function**: ReLU6 is utilized as the activation function, which is particularly advantageous for mobile and embedded platforms. ReLU6 is a variant of the Rectified Linear Unit (ReLU) activation function, defined as $\text{ReLU6}(x) = \min(\max(0,x), 6)$. The performance of the classification task is evaluated using categorical cross-entropy, ensuring that the model accurately differentiates between violent and non-violent videos.
- **Training Setup:** The model was initially developed and validated using a 30-70 split hold-out validation method. The model was trained with 700 videos and tested on the remaining 300 for each epoch.

Fig 2. below showcases the accuracy and loss trends during the model's training and validation phases, demonstrating the model's performance over time.

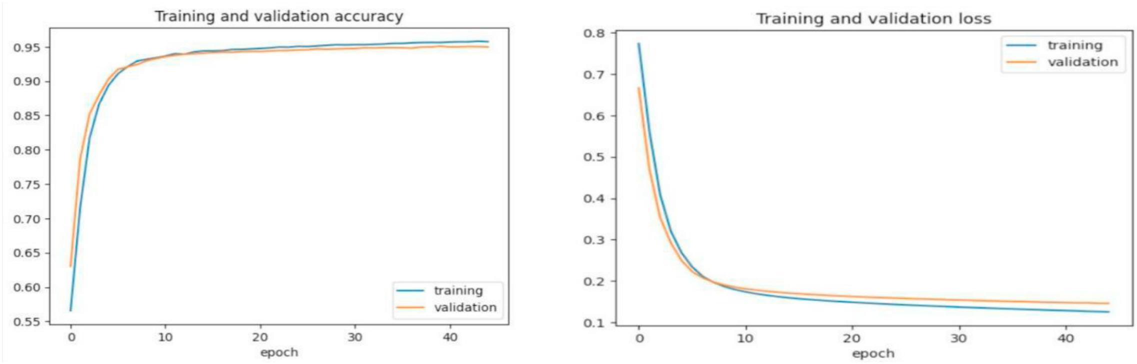

Fig 2. Accuracy and loss during the training and validation of the model.

3.3. Violence detection and alert module

Once the model is trained, video frames are processed sequentially to identify violent behavior. Spatial features such as edges, textures, and shapes are extracted by the model while temporal features are examined to understand motion patterns and contextual changes. Spatial features derived from each frame represent the static visual appearance of the video content, whereas temporal features encapsulate the possible motion information over time. This dual analysis enables efficient detection of violent actions by leveraging both static and temporal visual data. Frames are processed one at a time, and an alert is generated if violence is detected in any of them. The procedure for identifying a violent activity and generating an alert is demonstrated in **Algorithm 1**.

Algorithm 1: Alert Module for Real-Time Violence Detection System

```
begin
  1. Initialize counter to 0
  2. For each frame where violence is detected:
       If counter == 0:
           Set counter to 1
       Else:
           Increment counter by 1
       If counter == 30:
           Threshold reached
           Proceed to Step 3
  3. Collect frame data
       timestamp = get_current_timestamp()
       snapshot = capture_current_frame()
       location = get_camera_location()
       Proceed to Step 4
  4. Compose Alert Message   alert_message = compose_message(snapshot, timestamp, location)
  5. Send Alert   send_to_telegram(alert_message)
  6. Reset Counter   counter = 0
end
```

4. Evaluations

This section discusses the dataset used for evaluating the proposed system and then presents the evaluation results and observations. The system successfully distinguishes between violent and non-violent activities captured in surveillance footage. This section also presents the performance comparison of the proposed model with some models that exist in the literature.

Dataset: The system is evaluated on two datasets, Set 1 that is balanced [3] and Set 2 that is unbalanced [12]. The balanced dataset consists of 1000 video clips, evenly split between violent and non-violent content, with 500 clips in each category. This dataset features CCTV footage, providing valuable resource for analyzing and understanding patterns. The unbalanced dataset consists of 1000 video clips, where the majority are non-violent, and a smaller portion depicts violence mostly from CCTV footage. Each clip averages about 5 seconds in length.

Evaluation metrics: To comprehensively assess the model's performance, we employ several metrics, including accuracy, precision, recall, and F1-score. These metrics are crucial for understanding the effectiveness of the model in identifying violent behavior. The model is trained and evaluated using a 30-70 split hold-out validation method. The accuracy and loss of the proposed model is evaluated for training, validation and testing data at 40 epochs is shown in Table 1.

Table 1. Loss and Accuracy of the proposed model.

Set 1	Loss	Accuracy
Training	0.124	0.962
Validation	0.139	0.957
Testing	0.143	0.952

Set 2	Loss	Accuracy
Training	0.137	0.958
Validation	0.146	0.951
Testing	0.152	0.944

Additionally, the ROC curve is evaluated, which visually represents the trade-off between true positive rates and false positive rates. As shown in Fig. 3, the Area Under the Curve (AUC) of 0.99 signifies near-perfect performance in distinguishing between violent and non-violent activities, further confirming the model's strong classification ability.

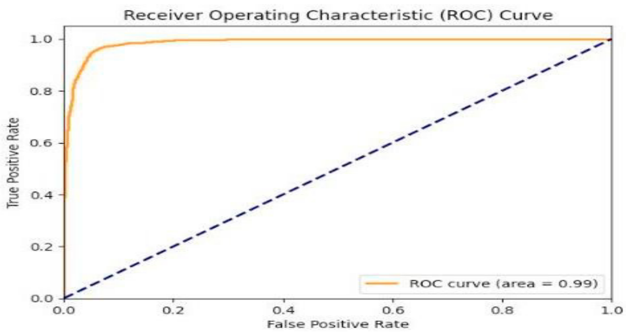

Fig 3. ROC curve result for the model

Comparison with other systems: The performance of our system is compared with existing models, and the evaluation results are summarized in Table 2. This comparison demonstrates that MobileNetV2 outperforms other architectures, including LSTM, ResNet50, and VGG16, across all key metrics (accuracy, precision, recall, and F1-score) on both balanced and unbalanced datasets. The results indicate that while the performance of the multimodal framework is comparable to our proposed system, it is also more computationally complex and time-consuming, making our approach more practical for real-time applications.

In comparison to the recent studies our system was able to achieve an accuracy of 95%, which demonstrates its potential in real-world scenarios. When compared with similar studies such as [5] and [6], our model showed an improvement in detection accuracy and a reduction in false positives. Unlike these studies, our model uses MobileNetV2 algorithm which allows for robust real-time detection in variable conditions. This demonstrates the effectiveness of our approach in practical world scenarios.

Table 2. Comparing the proposed system with the existing ones.

	Dataset Type	Accuracy	Precision (Violence/Non-Violence)	Recall (Violence/Non-Violence)	F1-Score (Violence/Non-Violence)
Proposed system	Set 1	0.95	0.94 / 0.94	0.95 / 0.94	0.94 / 0.93
	Set 2	0.94	0.92 / 0.92	0.94 / 0.92	0.93 / 0.92
LSTM [5]	Set 1	0.89	0.85 / 0.87	0.82 / 0.86	0.87 / 0.84
	Set 2	0.88	0.84 / 0.86	0.81 / 0.85	0.85 / 0.83
ResNet50 [6]	Set 1	0.93	0.92 / 0.91	0.91 / 0.90	0.92 / 0.91
	Set 2	0.92	0.91 / 0.89	0.89 / 0.88	0.90 / 0.88
Multimodal [4]	Set 1	0.94	0.93 / 0.93	0.92 / 0.91	0.92 / 0.92
	Set 2	0.93	0.92 / 0.91	0.91 / 0.90	0.91 / 0.91

5. **Conclusion**

The work is successfully developed and presented as an efficient Real-Time Violence Detection System utilizing the MobileNetV2 architecture. By analysing visual data from the surveillance footage, aggressive behaviour can be accurately detected in real time by the system, which sends immediate alerts to security personnel via the Telegram bot. Extensive evaluations demonstrate that the system's superior performance is achieved compared to existing models in distinguishing aggressive from non-violent actions. A groundbreaking solution is offered that leverages machine learning and computer vision to proactively identify and respond to violent incidents. By combining high accuracy, speed, and user-friendly alerts, a robust and scalable approach to enhancing public safety is provided.

While a strong foundation for real-time violence detection systems is laid by this study, further research is deemed essential to enhance its effectiveness. Future studies should focus on exploring larger and more diverse datasets to improve model robustness. Additionally, advanced deep learning techniques, such as ensemble methods or hybrid models, should be investigated, as they may yield better performance. Finally, user studies should be conducted to assess the system's real-world applicability, and the notification mechanisms can be refined to further optimize the system for practical deployment.

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